ETEFOP TSS FIRST TERM COMPREHENSIVE ASSESSMENT

SECTOR: ICT & Multimedia|TRADE: Networking and Internet Technologies | RQF LEVEL 5

ACADEMIC YEAR 2025–2026 | DATE: 11 December 2025

MODULE TITLE AND CODE: MACHINE LEARNING FUNDAMENTALS (NITML501)

DURATION: 3 Hours

MAXIMUM MARKS: 100 Marks

Trainer: MPAYIMANA CYIZA Landry

Trainee: _____

INSTRUCTIONS TO STUDENTS

- This paper contains Section A and Section B.
- All questions in Section A are compulsory (70 Marks).
- Attempt ANY THREE (3) questions in Section B (30 Marks).
- Write clearly; unclear answers score zero.

SECTION A (70 Marks) — ALL COMPULSORY

Q1. Machine Learning enables computers to learn patterns from _____ and improve automatically.

Q2. Which tool is commonly used in ML environment setup for running notebooks?
A. Adobe Illustrator
B. Jupyter Notebook
C. MS Word
D. FileZilla

Q3. A school gathers attendance logs for prediction. Which ML stage is this?
A. Data Acquisition
B. Model Deployment
C. Hyperparameter Tuning
D. Prediction Output

Q4. Data Cleaning helps remove errors, inconsistencies, and missing values before model building. (True/False)

Q5. Which visualization method helps identify clusters?
A. Line Chart
B. Histogram
C. Scatter Plot
D. Box Plot

Q6. Collecting raw information from sensors, APIs, or surveys is known as data _____.

Q7. Match the resource with category:
1. Laptop
2. Python
3. Dataset
A. Material
B. Tool
C. Equipment

Q8. A bank wants to visualize transaction amounts to identify anomalies. Which visualization is best?
A. Heatmap
B. Pie Chart
C. Word Cloud
D. Gauge Meter

Q9. Data Preparation occurs before Model Selection in ML lifecycle. (True/False)

Q10. Which is a valid data cleaning activity?
A. Adding random values
B. Removing duplicates
C. Increasing file size
D. Disabling logs

Q11. Representing data graphically to interpret trends is called data _____.

Q12. A researcher downloads climate data online. This is:
A. Data Wrangling
B. Data Visualization
C. Data Acquisition
D. Data Modeling

Q13. ML uses experience (data) to improve predictions over time without explicit programming. (True/False)

Q14. Match visualization types:
1. Histogram
2. Line Chart
3. Box Plot
   A. Trends over time
   B. Data distribution
   C. Spread and outliers

SECTION B (30 Marks) Attempt ANY THREE (3) Questions for each question A = 3 marks, B = 3 marks, C = 4 marks.

Q15. Dataset: TemperatureReadings.csv

Day	Temperature (°C)
1	22
2	24
3	21
4	26
5	25

A. State one purpose of data visualization in ML.
B. Identify an appropriate chart for this dataset and justify.
C. Describe how the visualization would appear and give one insight.

Q16. Dataset: ToClean.csv

ID	Age	Income	Missing?
1	25	35000	No
2	27	—	Yes
3	—	41000	Yes
4	30	39000	No
5	28	—	Yes

A. State one data cleaning activity.
B. Identify two missing data issues.
C. Explain two cleaning techniques and why they're important.

Q17. Dataset: ProductViews_vs_Purchases.csv

Product	Views	Purchases
A	120	10
B	80	6
C	150	20
D	60	2

A. Name one visualization technique for two numeric variables.
B. Select two variables to plot and justify.
C. Describe scatter plot pattern and interpret relationship.

Q18. Dataset: StudyHours_vs_Score.csv

Student	Hours	Score
S1	2	45
S2	4	60
S3	1	30
S4	5	75
S5	3	55

A. State one purpose of interpreting visualizations before modeling.
B. Identify one visible trend.
C. Explain how plotting supports ML data preparation.

Success!